RAJESH K. JAT

AI Engineer

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Professional Summary

AI Engineer with 2+ years of experience in computer vision, geospatial analytics, and edge ML optimization. Specialized in real-time video surveillance, satellite imagery analysis, and trajectory-based pattern recognition. Skilled in deploying CUDA-optimized models on NVIDIA Jetson and Linux environments using Docker and FastAPI for low-latency AI applications.

Core Competencies

- **Computer Vision:** Object detection, tracking, motion flow, pose estimation, OpenCV, FFmpeg.
- **Geospatial Analytics:** Raster data processing, QGIS, Rasterio, satellite imagery pipelines, real-world measurement.
- **Edge ML Optimization:** TensorRT quantization, CUDA kernels, NVIDIA Jetson deployment, real-time inference tuning.
- **Backend/API Deployment:** FastAPI, Docker, RESTful microservices, Linux server administration.
- **Applied AI:** Event-level metadata extraction, temporal correlation, trajectory analysis, model performance benchmarking.

Experience

AI Engineer Jul 2023 – Present

WESEE – Naval R&D Lab, New Delhi

- Designed and deployed a **satellite image analytics system** using YOLO + Oriented Bounding Boxes for maritime surveillance.
- Processed raster imagery with tiling, normalization, and coordinate scaling to improve inference performance and accuracy.
- Developed a **real-time surveillance analytics tool** for object tracking and motion detection, delivering explainable event logs under strict latency constraints.
- Optimized inference pipelines using TensorRT and CUDA, achieving sub-50 ms window latency on Jetson devices.
- Integrated FastAPI + Docker stack for modular deployment of computer vision services.

AI Intern Sep 2022 – Jan 2023

Object Automation System Solutions Pvt. Ltd., Chennai

- Developed and deployed a Flask-based recommendation engine on Linux VPS.
- Assisted in integrating computer vision modules for industrial automation prototypes.

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Projects

Flying Object Flow Tracker

Implemented motion analytics model to estimate trajectory, velocity, and encirclement using OpenCV-based frame differencing and vector math.

Sponsor Analytics in Sports
Developed YOLO-based system to detect brand logos in sports videos, capturing on-screen duration, ROI zones, and tracking frequency.

Satellite Object Detection
Built and deployed YOLOv8 model for warship and vessel identification using geospatial metadata for real-world dimension estimation.

Education

M.Tech in Artificial Intelligence & Data Science 2021 – 2023
IIIT Kota, MNIT Jaipur Campus

B.Tech in Computer Science & Engineering 2017 – 2021
Rajasthan Technical University

Certifications

- AWS Fundamentals – Coursera
- Developing Cloud Native Applications – Coursera

Technical Skills

- **Languages:** Python, C++, JavaScript, SQL
- **Libraries/Frameworks:** YOLO, OpenCV, TensorRT, PyTorch, TensorFlow, FastAPI, ReactJS, Rasterio, pyvips, OpenSeadragon, DeepStream
- **DevOps/Tools:** Docker, Git, GitLab (Self-hosted), NVIDIA CUDA, Linux Administration
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Platforms:** Ubuntu, AWS